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% MATLAB DC Machine Simulator % Author: Anjali % Institution: Bakhtiyarpur College of
Engineering (BCE Bakhtiyarpur) % Description: Numerical analysis and steady-state
characteristic plots % for DC Shunt and Separately Excited Motors. %
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clc; clear; close all;
%% 1. Machine Parameters (Default Values) V_supply = 230; % Supply Voltage (Volts) R_a =
0.5; % Armature Resistance (Ohms) R_f = 200; % Field Resistance (Ohms) K_phi = 1.2; %
Machine Constant * Flux (V-s/rad or N-m/A) T_load = 25; % Rated Load Torque (N-m)
%% 2. Operating Point Calculations fprintf('\n'); fprintf(' MATLAB DC MACHINE SIMULATOR
OUTPUT \n'); fprintf(' Author: Anjali | Bakhtiyarpur College of Engg. \n'); fprintf('\n\n');
% Field Current (Shunt Configuration) I_f = V_supply / R_f;
% Armature Current calculation for rated load torque % Torque T = K_phi * I_a ==> I_a = T /
K_phi I_a_rated = T_load / K_phi;
% Back EMF E_b = V_supply - (I_a_rated * R_a);
% Speed Calculations omega_rad_s = E_b / K_phi; % Speed in rad/s N_rpm = omega_rad_s *
(60 / (2*pi)); % Speed in RPM
% Power Calculations P_in = V_supply * (I_a_rated + I_f); % Total Input Power P_dev = E_b *
I_a_rated; % Developed Mechanical Power Efficiency = (P_dev / P_in) * 100;
% Display Computed Steady-State Values fprintf('Operating Parameters at T_load = %.2f
N-m:\n', T_load); fprintf(' - Field Current (I_f) : %.2f A\n', I_f); fprintf(' - Armature Current (I_a) :
%.2f A\n', I_a_rated); fprintf(' - Back EMF (E_b) : %.2f V\n', E_b); fprintf(' - Motor Speed : %.2f
RPM (%.2f rad/s)\n', N_rpm, omega_rad_s); fprintf(' - Power Output (Dev.) : %.2f W\n', P_dev);
fprintf(' - Overall Efficiency : %.2f %%\n', Efficiency);
fprintf('===== \n');
%% 3. Parametric Analysis & Visualization I_a_vec = linspace(0, 50, 200); % Armature Current
vector (0 to 50 A)
% Equations across current range E_b_vec = V_supply - (I_a_vec .* R_a); N_vec_rpm =
(E_b_vec ./ K_phi) .* (60 / (2*pi)); Torque_vec = K_phi .* I_a_vec;
P_in_vec = V_supply .* (I_a_vec + I_f); P_out_vec = E_b_vec .* I_a_vec; Eff_vec = (P_out_vec
./ P_in_vec) .* 100; Eff_vec(I_a_vec == 0) = 0; % Handle division by zero at startup
%% 4. Plotting Characteristics figure('Name', 'DC Machine Simulator - Characteristics',
'NumberTitle', 'off');
% Plot 1: Torque vs Speed subplot(2, 2, 1); plot(Torque_vec, N_vec_rpm, 'b-', 'LineWidth', 2);
grid on; title('Speed vs. Torque Characteristic'); xlabel('Electromagnetic Torque T (N-m)');
ylabel('Speed N (RPM)'); xline(T_load, 'r--', 'Rated Torque');
% Plot 2: Armature Current vs Torque subplot(2, 2, 2); plot(I_a_vec, Torque_vec, 'm-',
'LineWidth', 2); grid on; title('Torque vs. Armature Current'); xlabel('Armature Current I_a (A)');
ylabel('Torque T (N-m)');
% Plot 3: Speed vs Armature Current subplot(2, 2, 3); plot(I_a_vec, N_vec_rpm, 'g-', 'LineWidth',
2); grid on; title('Speed vs. Armature Current'); xlabel('Armature Current I_a (A)'); ylabel('Speed
N (RPM)');
% Plot 4: Efficiency vs Developed Power subplot(2, 2, 4); plot(P_out_vec/1000, Eff_vec, 'k-',
'LineWidth', 2); grid on; title('Efficiency vs. Power Output'); xlabel('Developed Power (kW)');
ylabel('Efficiency (%)');
sgtitle('DC Shunt Motor Performance Curves | Anjali (BCE Bakhtiyarpur)');

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